

Recurrences, Master's Theorem, Substitution Method

(Algorithm Design and Analysis (ADA) - CSE222)

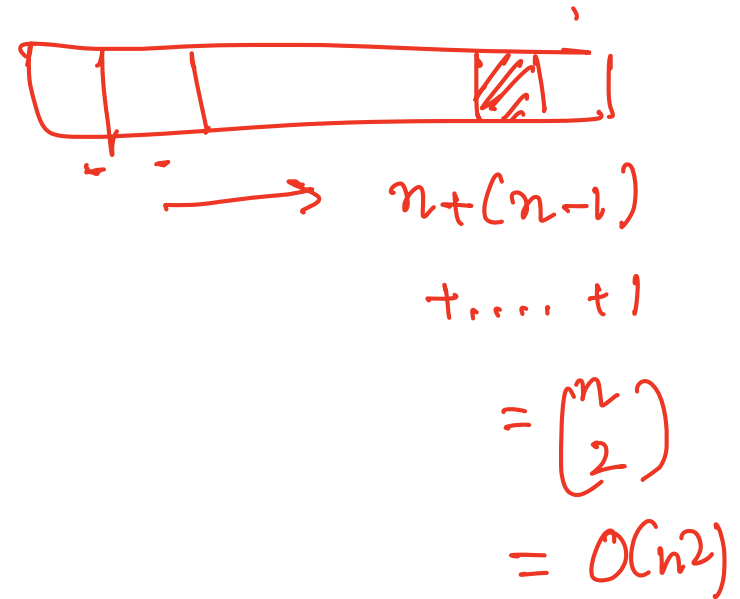
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January 19, 2023

Outline

- ➊ What have we seen in DSA?
- ➋ Master's Theorem
- ➌ Substitution Method
- ➍ References

- **Sorting Algorithms:** Bubble Sort, Insertion Sort, Quick Sort, Merge Sort.
- Bubble-sort and Insertion Sort both need $\mathcal{O}(n^2)$ -time.
- Can we do better?



$$\begin{aligned}
 &\rightarrow n + (n-1) \\
 &\quad + \dots + 1 \\
 &= \binom{n}{2} \\
 &= \mathcal{O}(n^2)
 \end{aligned}$$

- **Sorting Algorithms:** Bubble Sort, Insertion Sort, Quick Sort, Merge Sort.
- Bubble-sort and Insertion Sort both need $\mathcal{O}(n^2)$ -time.
- Can we do better?
- Answer is yes – merge-sort.

- **Input:** An array A of n numbers.
- **Output:** Sort the array A .

$$T(n) = aT\left(\frac{n}{b}\right) + n^d$$

$$T(1) = 1$$

Basic Steps of Merge Sort:

- If $n = 1$, then return A .
- $A_L = A(1, \dots, \lfloor n/2 \rfloor)$ and $A_R = A(\lfloor n/2 \rfloor + 1, \dots, n)$.
- Recursively sort A_L and A_R .
- $A = \text{Merge}(A_L, A_R) \rightarrow 2n$
- Output A .



$$T(n) = 2T\left(\frac{n}{2}\right) + 2n \quad \text{for } n \geq 2$$

$$T(1) = 1$$

$$O(n \log n)$$

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Master's Theorem

- **What we prove here:** If $T(n) = aT(n/b) + \alpha n^d$ and $T(1) = \beta$ (α and β are constants), then

Case 1: If $d = \log_b a$, then $T(n) = \mathcal{O}(n^d \log_2 n)$.

Case 2: If $d < \log_b a$, then $T(n) = \mathcal{O}(n^{\log_b a})$.

Case 3: If $d > \log_b a$, then $T(n) = \mathcal{O}(n^d)$.

Assume
 n is a
power of b

- **Actual statement:** If $T(n) = aT(n/b) + \Theta(n^d)$ and $T(1) = \Theta(1)$, then

Case 1: If $d = \log_b a$, then $T(n) = \Theta(n^d \log_2 n)$.

Case 2: If $d < \log_b a$, then $T(n) = \Theta(n^{\log_b a})$.

Case 3: If $d > \log_b a$, then $T(n) = \Theta(n^d)$.

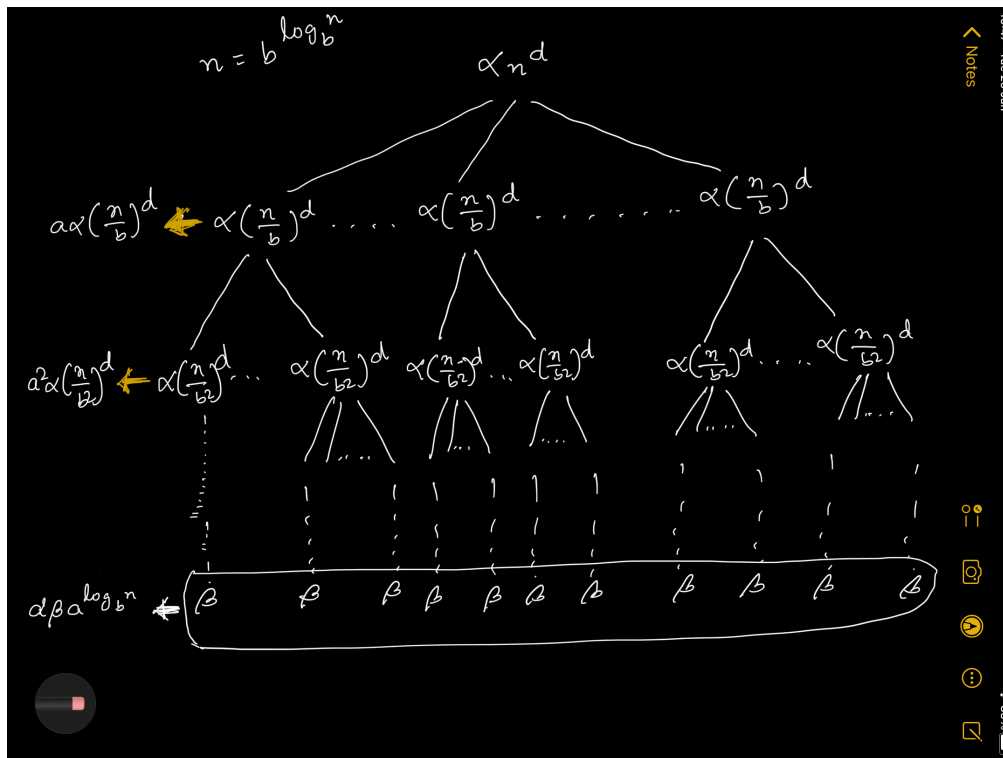
Proof (Sketch)

- $T(n) = aT(n/b) + \alpha n^d$ and $T(1) = \beta$ (α and β are constants)

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- $T(n) = aT(n/b) + \alpha n^d$ and $T(1) = \beta$ (α and β are constants)

$$n = b^{\log_b n}$$



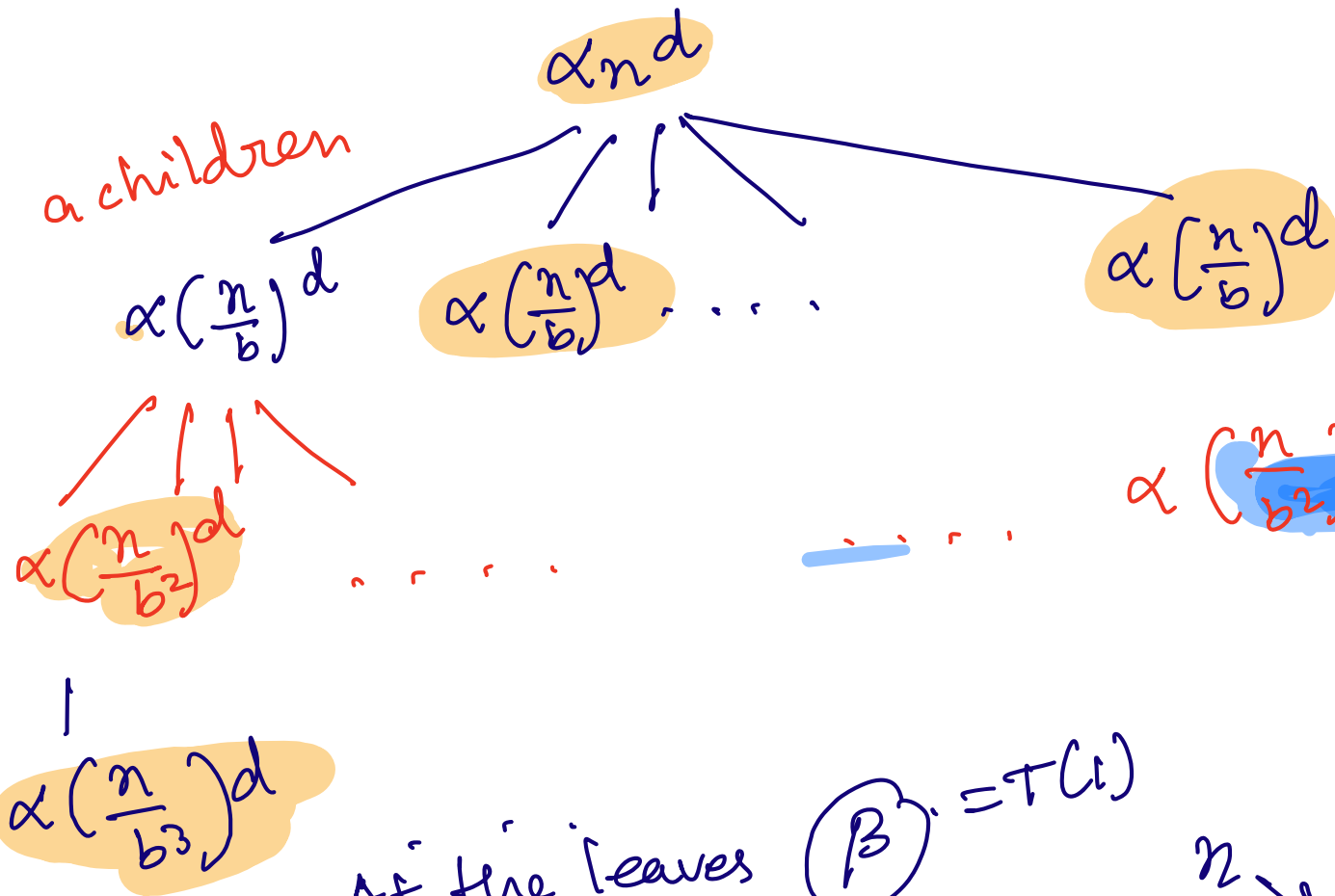
- The value at every node represents the time taken (or the number of steps) or cost at each level in recursion.
- Total number of leaves is $a^{\log_b n} = n^{\log_b a}$.
- The cost of computation (or the number of steps at the leaves) is $\beta n^{\log_b a}$.
- Total cost of computation (number of steps) is $\beta n^{\log_b a} + \alpha \sum_{j=0}^{\log_b n - 1} a^j (n/b^j)^d$.

$$T(n) = aT\left(\frac{n}{b}\right) + \alpha n^d \text{ and } T(1) = \beta.$$

Recursion Tree

$\log_b n$
= depth of
recursion
tree
→ a times

$$\begin{aligned} \log_b n &= \log_b n \\ \log_b n &= \log_b n \end{aligned}$$



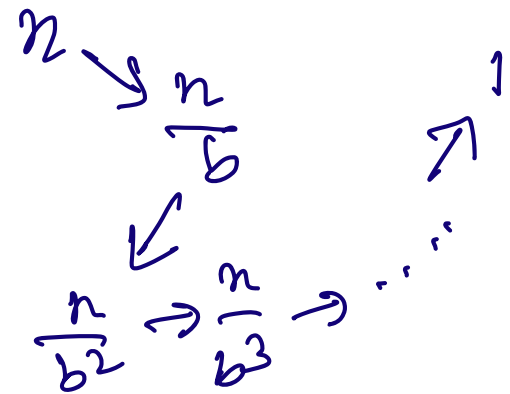
$$\alpha \left(\frac{n}{b^2}\right)^d \rightarrow a^2 \text{ times}$$

$a^3 \text{ times}$

At the leaves $\beta = T(1)$

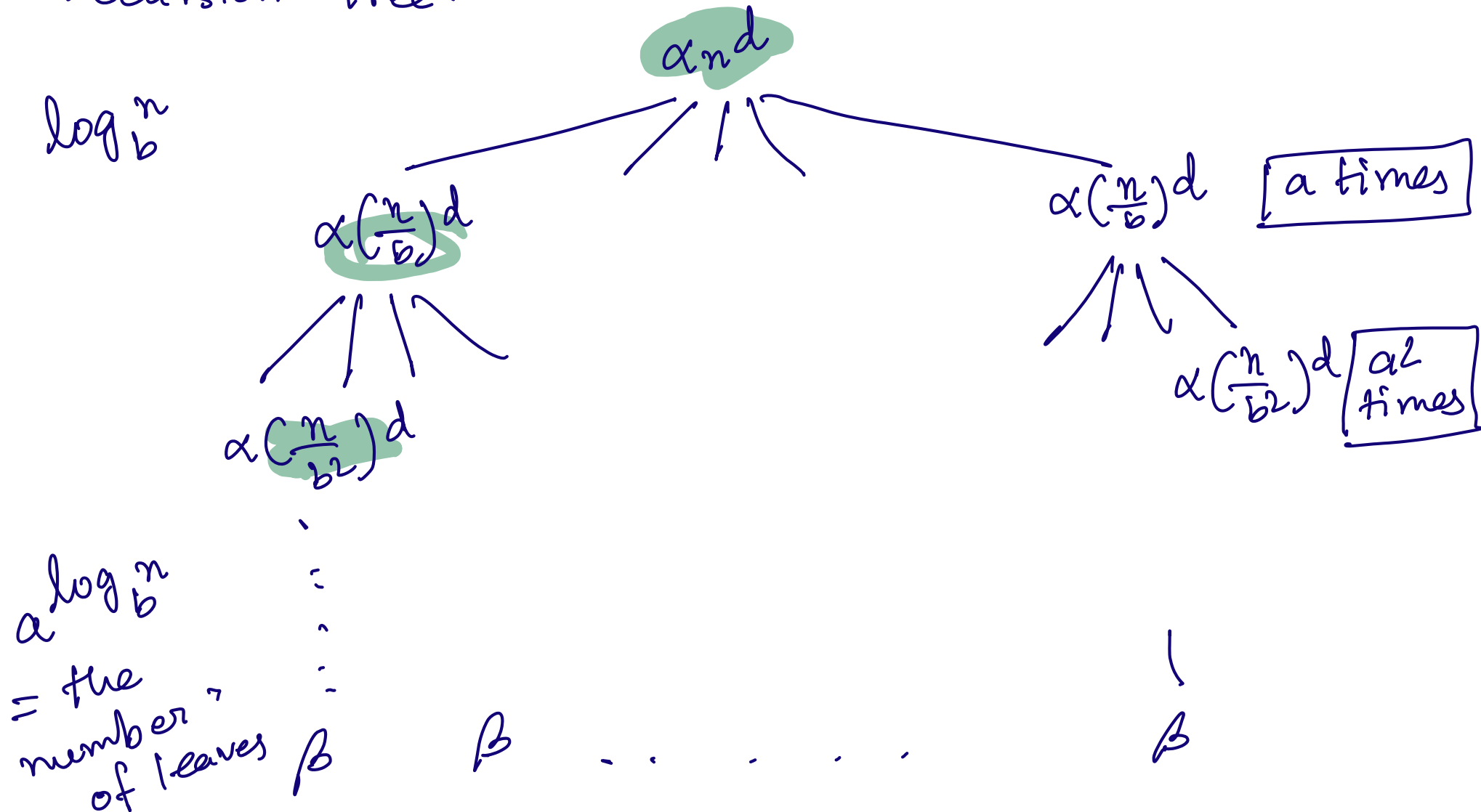
$\log_b n$ = depth of the recursion tree.

At the leaves $n=1$



Proof Sketch (Case 1)

$T(n)$ = The sum of the values of the nodes in the recursion tree.



Proof Sketch (Case 1)

$$T(n) = \alpha n^d + \alpha \left(\frac{n}{b}\right)^d + \alpha^2 \left(\frac{n}{b^2}\right)^d + \dots + \alpha^{\log_b n - 1} \left(\frac{n}{b^{\log_b n - 1}}\right)^d$$

$$d = \log_b a$$

$$= \beta n^{\log_b a} + \alpha \sum_{j=0}^{\log_b n - 1} a^j \left(\frac{n}{b^{j+1}}\right)^d + \beta \alpha^{\log_b n}$$

$$= \beta n^d + n^d \alpha \sum_{j=0}^{\log_b n - 1} a^j \left(\frac{1}{b^{j+1}}\right)^{\log_b a}$$

$$= \beta n^d + n^d \alpha \sum_{j=0}^{\log_b n - 1} a^j \left(\frac{1}{a^{j+1}}\right)$$

$$\begin{aligned} (b^j)^{\log_b a} &= (b^{\log_b a})^j \\ &= a^j \end{aligned}$$

Proof Sketch (Case 1)

$$= \beta n^d + n^d \cdot \alpha \cdot \log_b n$$

$$\approx \Theta(n^d \log_b n).$$

- **Actual statement:** If $T(n) = aT(n/b) + \Theta(n^d)$ and $T(1) = \Theta(1)$, then

Case 1: If $d = \log_b a$, then $T(n) = \Theta(n^d \log_2 n)$.

Case 2: If $d < \log_b a$, then $T(n) = \Theta(n^{\log_b a})$.

Case 3: If $d > \log_b a$, then $T(n) = \Theta(n^d)$.

α and β are constants.

d and $\log_b a$

$b \rightarrow$ rate of shrinking of subproblem

$a \rightarrow$ number of subproblems

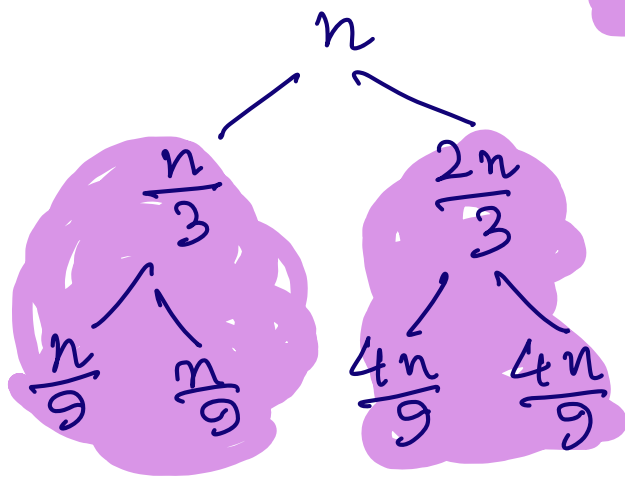
$$\log_b n \approx \Theta(\log_2 n)$$

$$\log |E| \approx O(\log n)$$

$$T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + n$$

Exercise.

$$\Theta(n \log_2 n)$$



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$$T(n) = T(n-1) + n.$$

$$T(1) = 1$$

$$\text{Guess } T(n) \leq cn^2$$

c is constant independent of n

→ Prove by induction

Base Case: $n=1$. $T(1)=1 \leq c^2$ for all $c \geq 1$

Induction Hypothesis: $n \leq k$, the guess holds true

$$T(k) \leq ck^2$$

Induction Step: $n = k+1$. $\leq c(k+1)^2?$

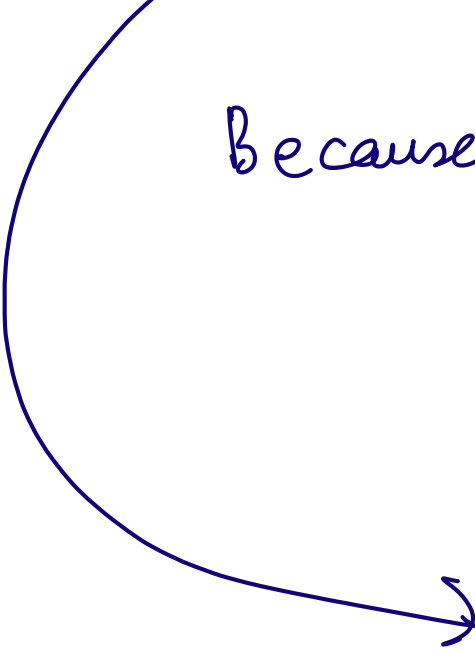
$$T(k+1) = T(k) + k+1$$

$$\leq ck^2 + k+1 \quad (\text{by induction hypothesis})$$

$$\leq ck^2 + 2ck + c \quad \text{for all } c \geq 1$$

Because $k \leq 2ck$ for any $c \geq 1$

$$1 \leq c$$


$$= c(k+1)^2$$

$$T(n) = O(n^2)$$

$$T(n) = 4T(n/2) + n.$$

$$\boxed{T(1) = 1}$$

↓

Base Case

By master's theorem $T(n)$ is $\Theta(n^2)$.

$$T(n) = 4T(n/2) + n.$$

Base Case: $T(1) = 0$

Guess 1: $T(n) \leq cn^2$.

Base Case: $n = 1$. Guess holds true $T(1) = 1 \leq cn^2$

Induction Hypothesis: Guess holds true for all values

$$n \leq k.$$

$$T(k/2) \leq c\left(\frac{k}{2}\right)^2 = ck^2/4.$$

$$T(k) = 4T\left(\frac{k}{2}\right) + k$$

$$\leq 4c\frac{k^2}{4} + k = ck^2 + k$$

Even a correct guess can lead to dead ends.

$$T(n) = 4T\left(\frac{n}{2}\right) + n$$

Guess 2: $T(n) \leq cn^2 - cn$.

Base Case: $T(1) = 0 \leq 0$ Holds true.

Induction Hypothesis: The guess holds true

for all values $n \leq k-1$ *(stronger mathematical induction)*

$$T\left(\frac{k}{2}\right) \leq c\left(\frac{k}{2}\right)^2 - c\left(\frac{k}{2}\right) = \frac{ck^2}{4} - \frac{ck}{2}$$

Induction Step: $n = k$.

$$T(k) = 4T\left(\frac{k}{2}\right) + k$$

$$\leq 4 \frac{ck^2}{4} - 4 \frac{ck}{2} + k$$

$$ck^2 - ck$$

$$= ck^2 - 2ck + k$$

$$= ck^2 - (2c - 1)k$$

Holds true for all $c \geq 1$

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Office Hours:

Wednesday 2-3 pm
B-501 R&D

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- Chapter 4 - Section 4.3, 4.4, and 4.5 of CLRS book.

Cormen, Lieresen, Rivest, Stein